ICB response prediction

Nemotron clinical copilot

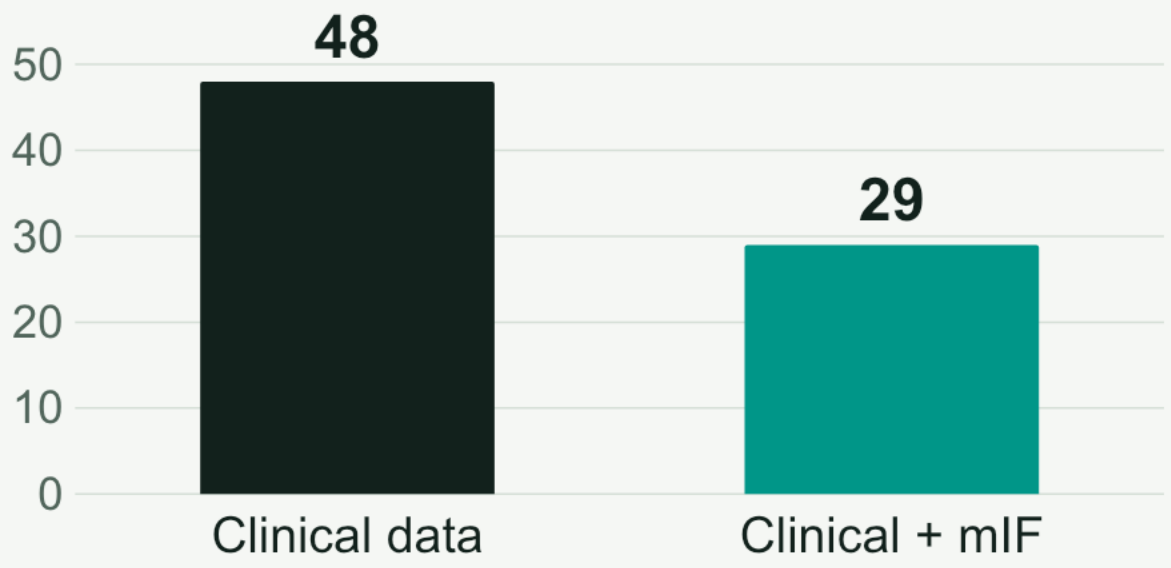
Execution pipeline and model harnesses

Research demonstration



Research scope

Available patients



Research question

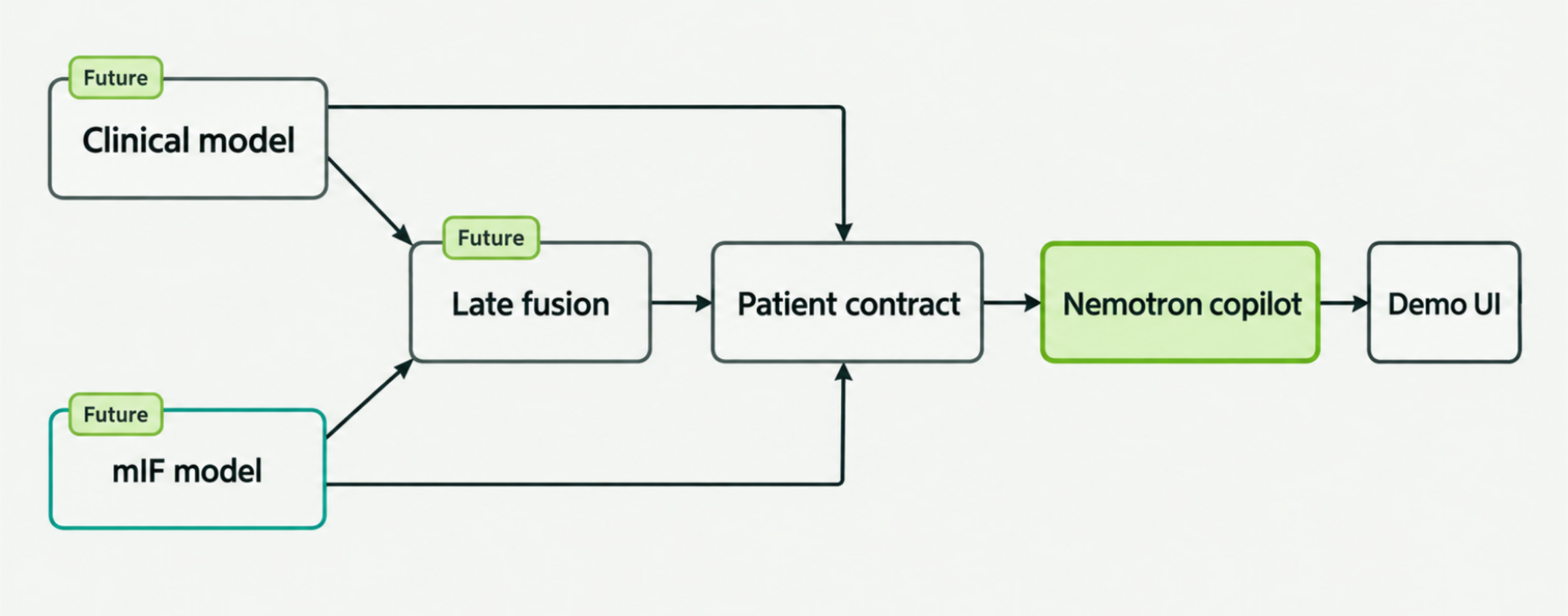
Does combining clinical and mIF information improve prediction of ICB responders?

Planned comparison

Clinical-only, mIF-only and late-fusion predictions on the same evaluation patients.

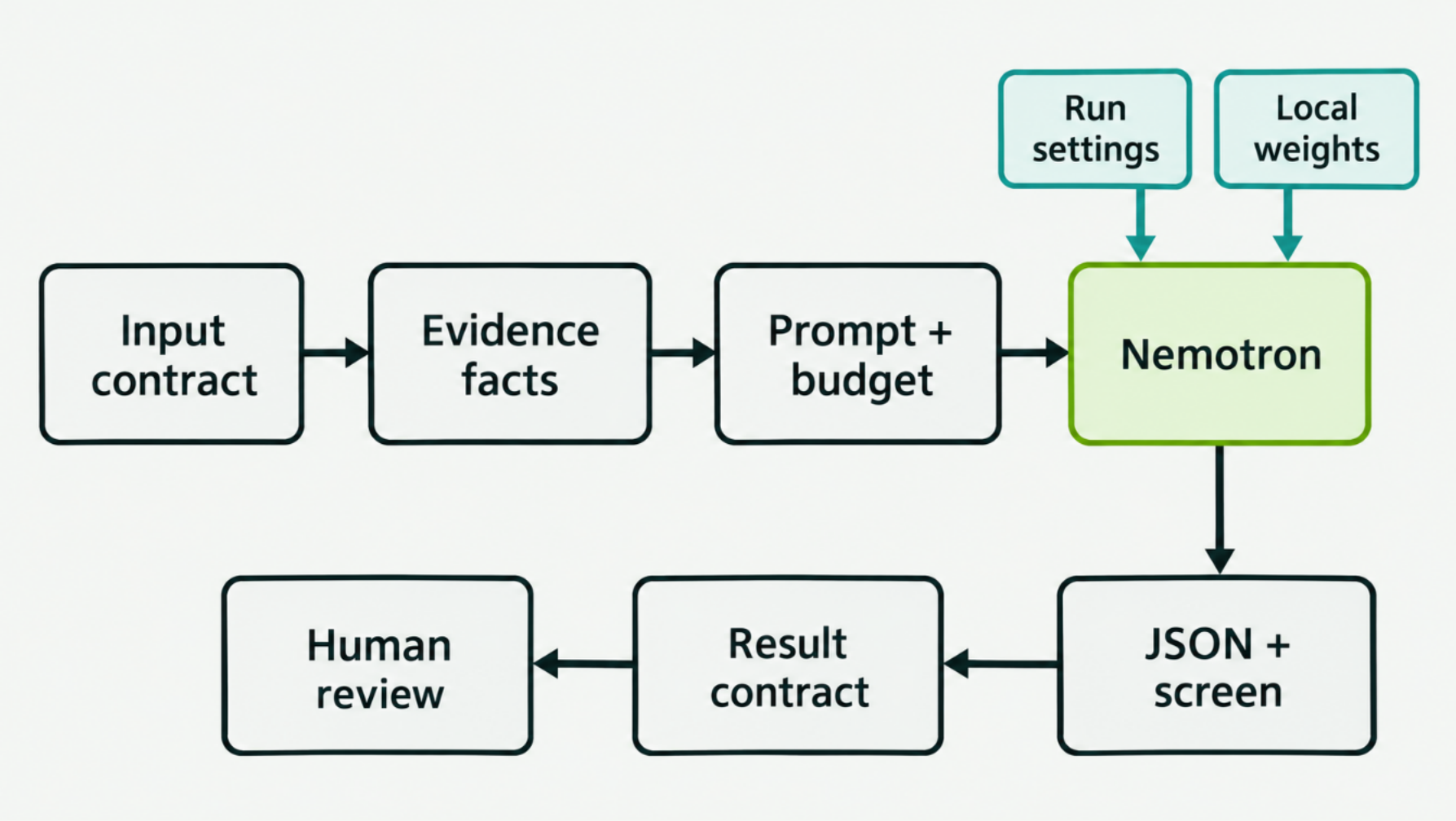
The 29 paired patients are a subset of 48. H&E and extra mIF batches remain possible extensions.

System architecture



Models export scores, features and context. This repository implements the copilot and its interfaces.

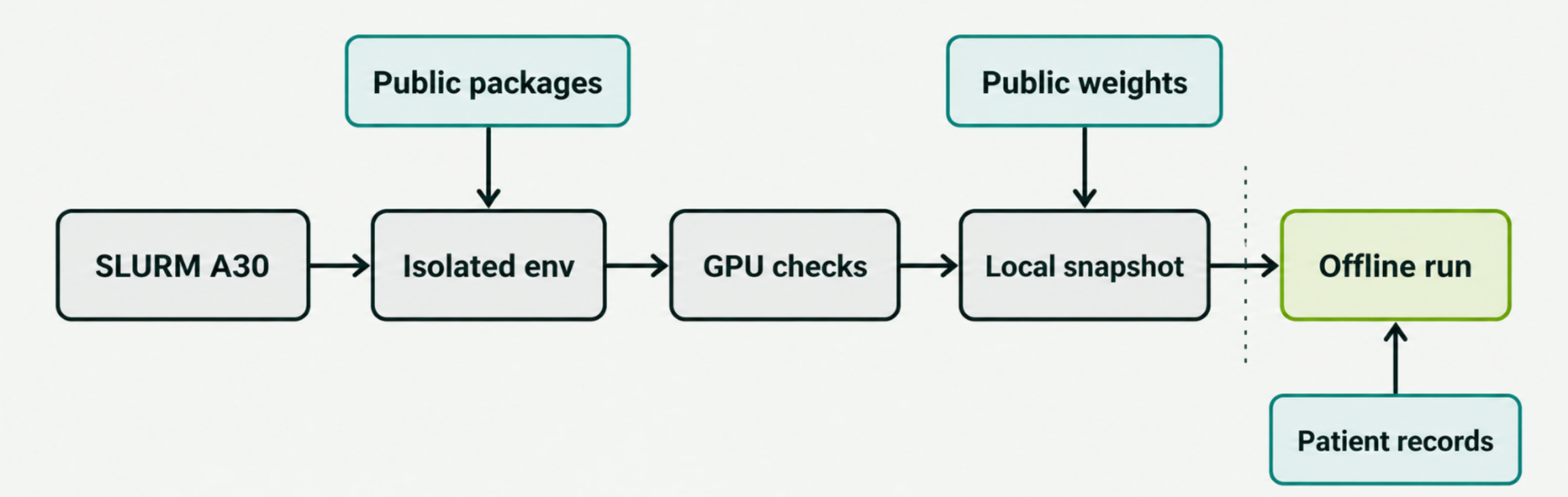
Application harness around Nemotron



The harness controls evidence and execution. Every summary requires independent human review.

HPC execution pipeline

```
bash run_hpc.sh
```



Preparation downloads public artifacts. Patient inference then uses local weights with offline settings.

The GPU engine loads lazily on the first interpretation request.

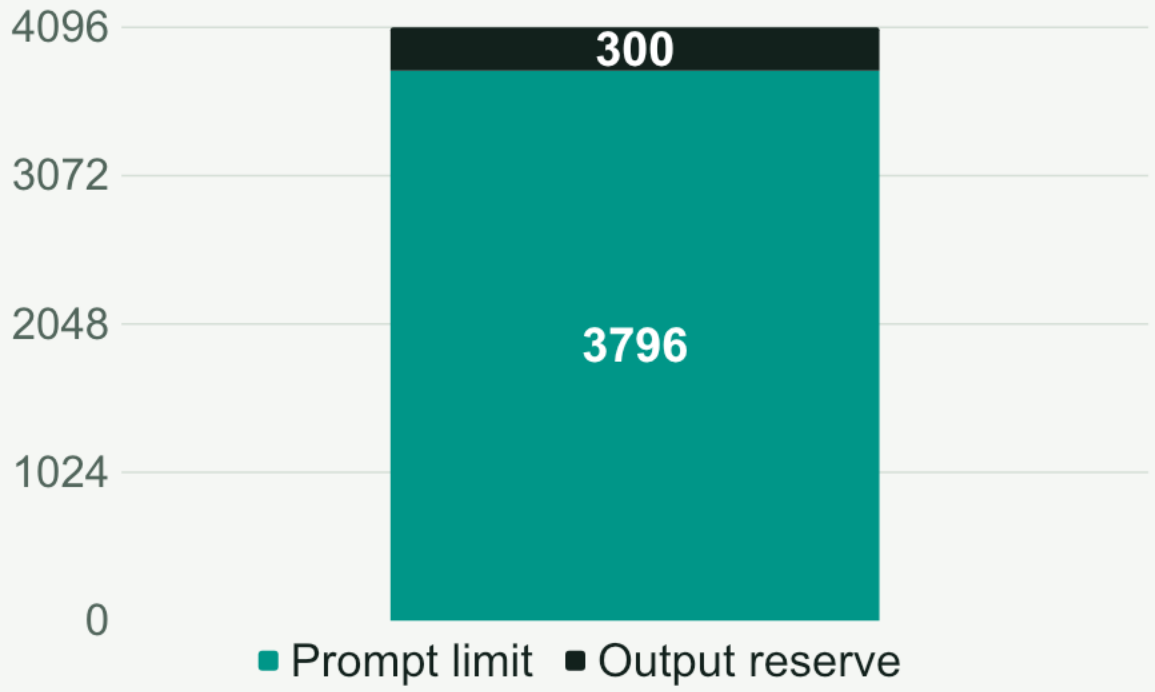
Runtime prerequisites

Layer	Selected deployment requirement
GPU	One A30 with 24 GB VRAM and BF16 support.
Host platform	Linux x86_64 with glibc 2.28 or newer.
NVIDIA driver	570.26 or newer for the selected native CUDA 12.8 path.
Python	Fresh Python 3.11 environment from the Anaconda module.
GPU libraries	vLLM 0.10.2 and PyTorch 2.8.0+cu128.
Tokenizer libraries	Transformers 4.55.2 and Hugging Face Hub 0.34.4.
Model snapshot	Local configuration, tokenizer, weight files and revision metadata.
Job defaults	32 GB host RAM, four CPUs and a four-hour allocation.

Doctor checks CUDA allocation. A generation experiment must verify model loading.

Token budget and execution settings

Default context allocation



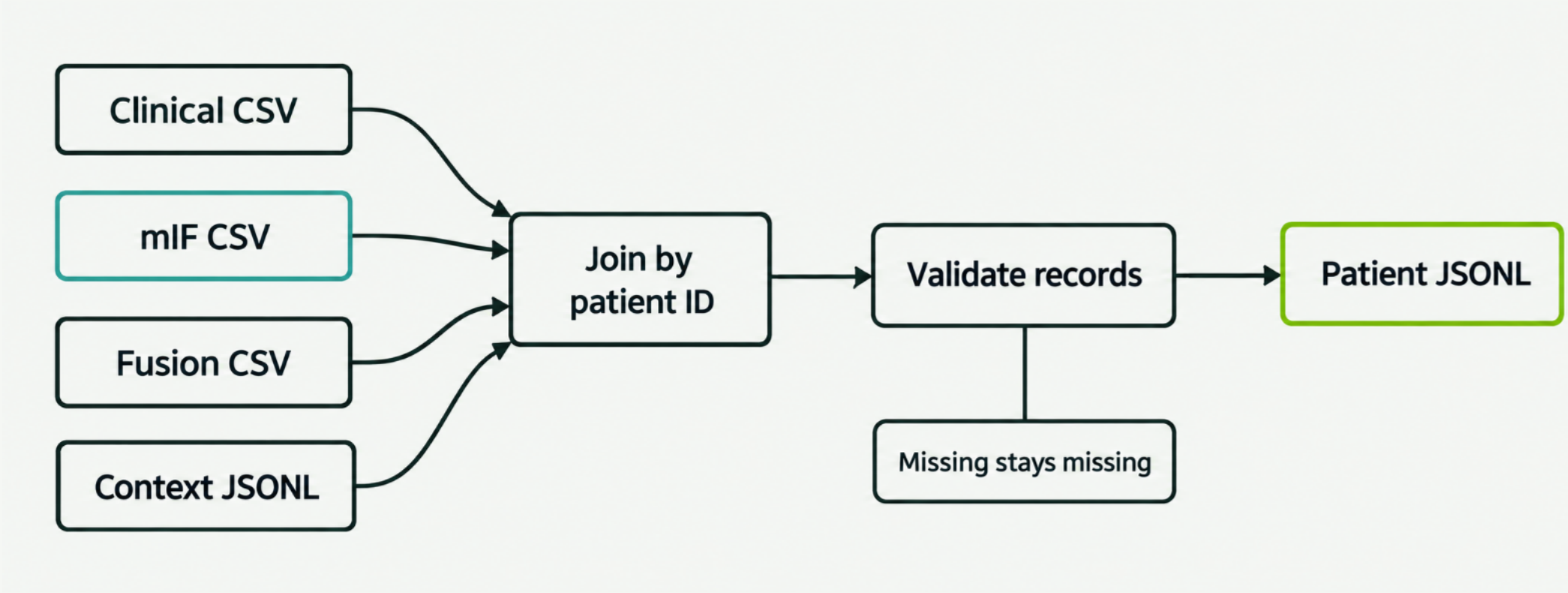
4,096 − 300 = 3,796 input tokens maximum

Current execution defaults

```
dtype = bfloat16
GPU memory fraction = 0.80
GPU concurrent sequences = 4
Experiment batch size = 8
temperature = 0.0
seed = 42
```

Character limits do not guarantee a prompt fits the token budget.

Patient joins and missing modalities



Identity gate

Resolve slide IDs upstream. Export one score per patient per modality.

Availability gate

Missing predictions remain absent. The structured result names unavailable modalities.

Shared patient contract

```
{
  "schema_version": "1.0",
  "patient_id": "P12",
  "predictions": {
    "clinical": {"probability": 0.21},
    "mif": {"probability": 0.78},
    "fused": {"probability": 0.57}
  },
  "response_definition": "Not specified"
}
```

Patient identity

One canonical patient ID and one prediction per patient per modality.

Score metadata

Model version, calibration status and optional threshold travel with each score.

Feature evidence

Measurement direction and signed attribution are separate.

Positive class: ICB responder. The team has not yet agreed the responder definition.

Feature evidence contract

```
{
  "feature": "LDH level",
  "value": 510,
  "unit": "U/L",
  "direction": "elevated",
  "attribution": {
    "method": "shap",
    "value": -0.18,
    "scale": "log_odds"
  }
}
```

Measurement meaning

Value, units and reference direction describe the supplied observation.

Model effect

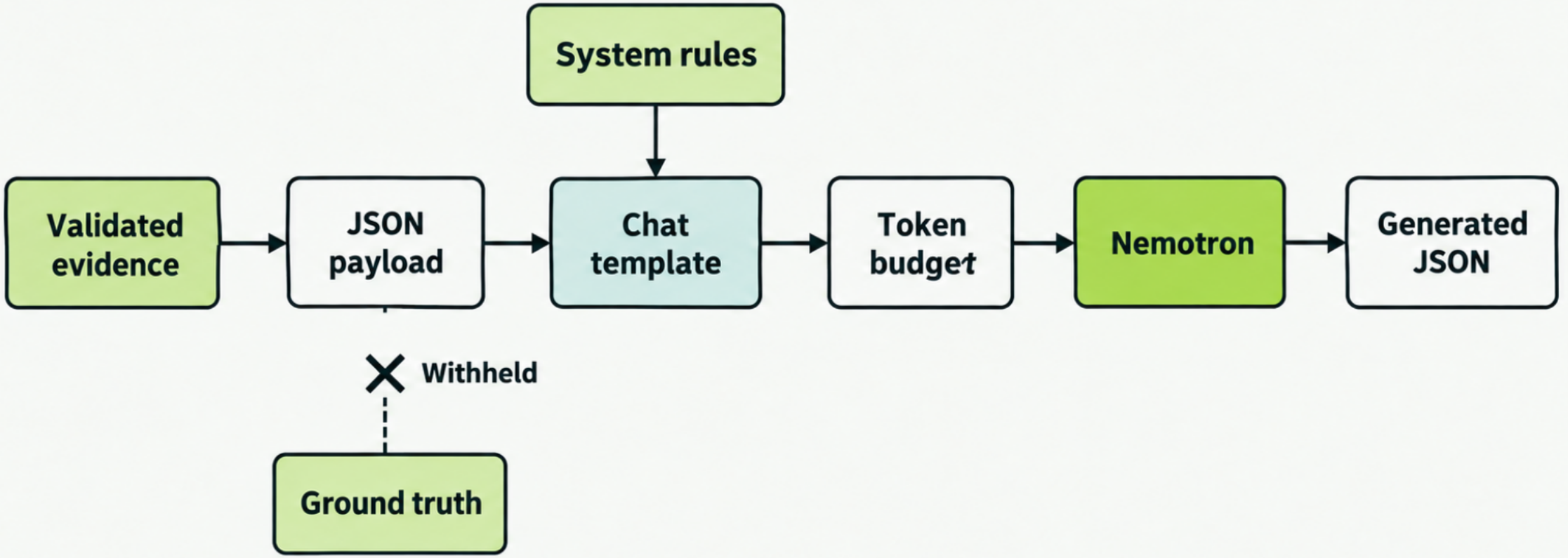
A signed attribution describes that model's responder score on a declared scale.

Upstream semantic checks

Confirm class mapping, units and prediction timepoint. Avoid causal claims.

Synthetic example. A high measurement alone does not determine the feature's effect on a model score.

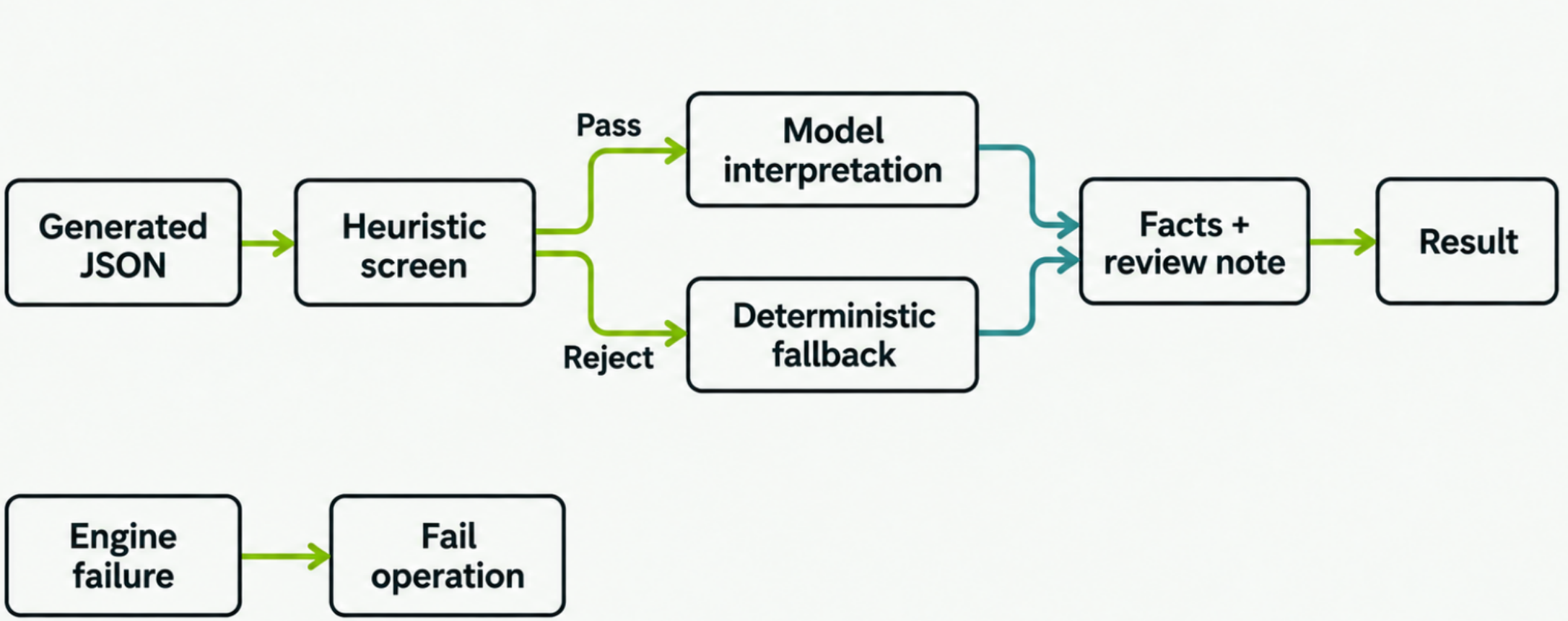
Prompt and context harness




```
Expected model JSON: {"interpretation": "Two concise research sentences."}
```

Ground truth stays outside generation. Human review still checks factual correctness and context leakage.

Output quality gates and fallback



Rejected prose falls back. Engine errors stop a batch run or return an API error.

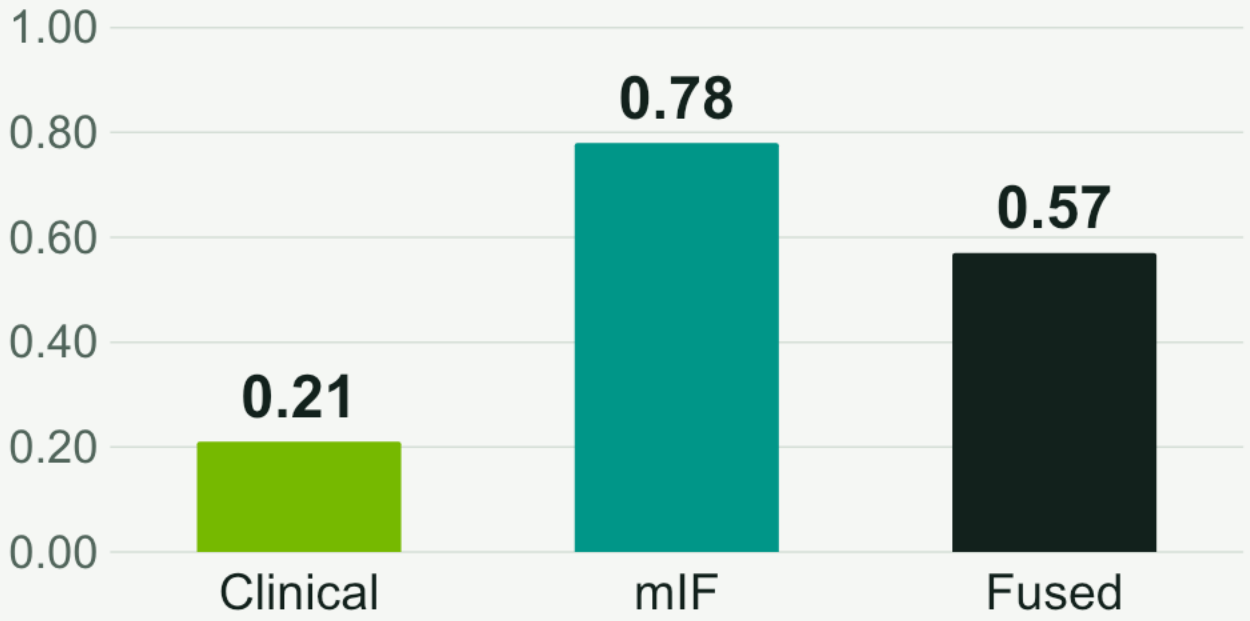
Expected result contract

Field group	Expected behavior
Identity and predictions	Same patient ID and validated scores. No fabricated missing score.
Narrative	Facts from input, a screened interpretation or fallback, and a review note.
Source and quality	Source: mock / vllm / fallback. Quality: mock / passed / fallback.
Comparison and limitations	Show availability, score gaps and supplied-threshold classifications.
Review status	review_required is always true. The summary remains research output.
Provenance	Record prompt version, backend, local model metadata and settings.

Result fields are implemented. A formal response-schema validator is a proposed hardening step.

Synthetic case: discordant predictions

P12 / synthetic model scores



“Clinical and spatial mIF predictions are discordant and warrant review of the supplied evidence.”

Clinical–mIF gap: 0.57
Calibration: unknown
Human review: required

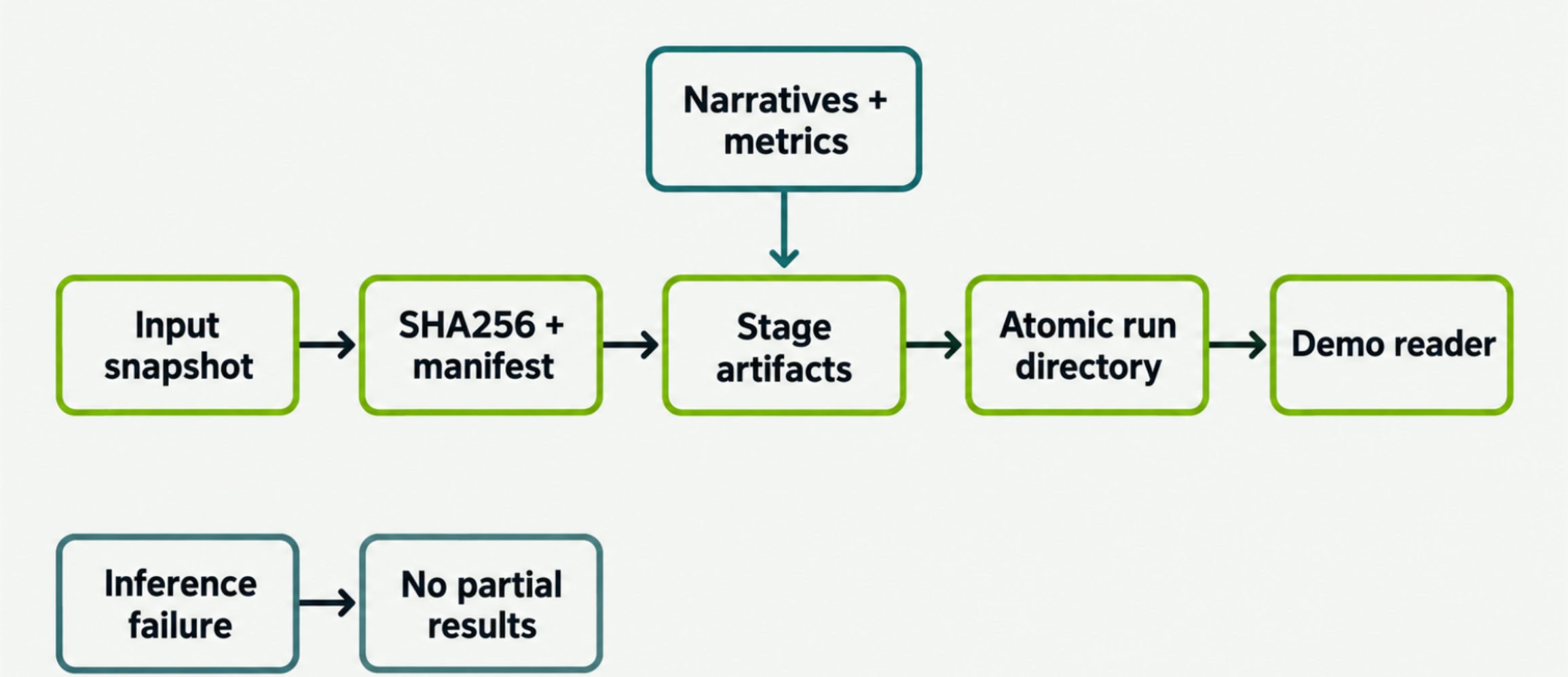
This is deterministic mock output. The supplied fused score does not establish better performance.

Failure handling and recovery

Failure	Current behavior	Operator response
Contract error	Reject input before generation.	Correct IDs, types or metadata.
Prompt too long	Reject before token generation.	Reduce approved context or features.
GPU / cache failure	Stop runtime or scheduled job.	Inspect doctor, cache and SLURM logs.
Rejected model prose	Publish deterministic fallback.	Inspect screen issues and review the case.
Engine failure	Stop run. API returns an error.	Fix the runtime and retry in a new run dir.
Existing result directory	Protect the completed run.	Choose a new output directory.

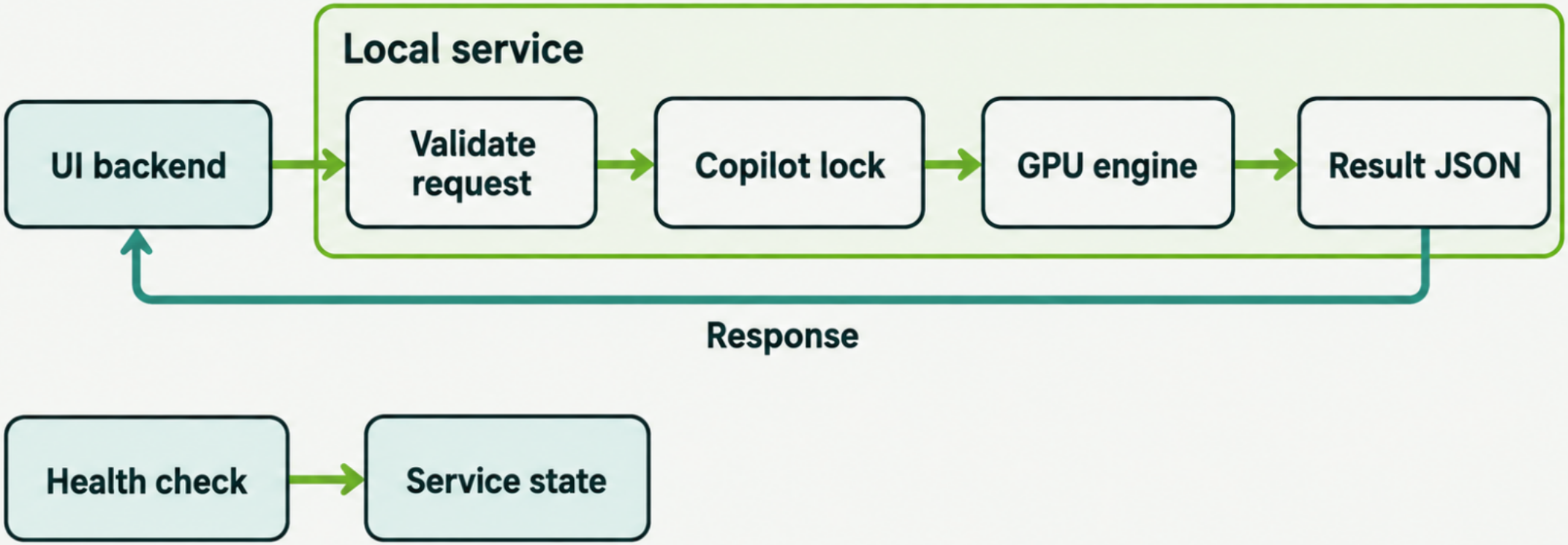
A fallback is a labeled summary. A runtime failure is an execution error, with no partial batch publication.

Experiment publication and provenance



narratives.jsonl narratives.csv report.md manifest.json metrics.json

Local API execution envelope



Current request limits

2 MB requests. Batches contain 1–128 patients. Model calls serialize.

Readiness distinction

Health checks service state. Generation verifies model execution.

Authentication, bounded queues and inference deadlines are proposed additions, not current API features.

Testing harness and readiness gates



Implemented tests

Schema, adapters, mocks, artifacts, HTTP and mocked SLURM tests pass.

Remaining site checks

Run four synthetic cases on A30. Review each resulting narrative.

Software tests verify the harness. Prediction and narrative quality need separate evidence.

Acceptance contracts for the real-data run

Gate	Evidence required before demo use
Identity and completeness	Expected IDs and row counts match. Missing modalities stay explicit.
Prediction meaning	Responder class, endpoint, units and attribution scales are confirmed.
Runtime execution	Real A30 generation succeeds and source/fallback counts are visible.
Narrative review	Review numeric facts, supplied evidence and unsupported claims.
Model evaluation	Compare eligible paired patients using agreed folds and endpoint.
Demo display	Show scores, source, limitations and the human-review flag.

Proposed acceptance rubric. The team must agree fallback tolerance, latency targets and the reviewer.

Connections to future models

Owner / workstream	Handoff to the copilot
Isis / clinical model	Clinical prediction CSV plus clinical features and context.
Alyssa / mIF model	mIF prediction CSV plus feature metadata and canonical patient IDs.
Fusion owner / TBD	Fused prediction CSV with model provenance for paired patients.
Paulo / copilot	Assemble, validate and interpret the combined patient records.
Junhao / demo UI	Read structured narrative results or call the local interpretation API.

Future H&E predictions require a versioned contract and adapter change. Raw images stay upstream.

Evaluation and review

Prediction quality

Use the same eligible patients across modalities.

Agree the endpoint and preserve patient-level folds.

Inspect AUROC and Brier score on paired outputs.

Narrative quality

Check reproduced scores and supplied features.

Review disagreement and missing modalities.

Inspect unsupported claims, source and limitations.

A successful mock run establishes interface behavior. Clinical validity and multimodal benefit require further evidence.

Harness hardening still to add

Proposed additions beyond the current implementation

Harness area	Addition to design and implement
Machine contracts	Versioned JSON Schemas and output cross-field invariants.
Evidence semantics	Check endpoint, timepoint, units, class mapping and fold provenance.
Live service controls	Model warmup, bounded queue, inference deadline and access policy.
Local observability	Request IDs, stage latency, live fallback counters and failed-run records.
Reproducibility	Hash-locked environment and weight digests when needed.
Acceptance automation	GPU regression cases plus independent narrative review.

These tasks strengthen execution and auditability. They do not establish clinical validity by themselves.

Decisions needed for the real-data demo

Proposed starting choices and owners for team confirmation

Decision	Suggested owner	Starting choice to confirm
Responder endpoint	Clinical owner TBD + Isis	Agree the responder definition and assessment window.
Patient identity	Isis + Alyssa	Use canonical IDs. Resolve slide aggregation upstream.
Evaluation design	Modeling team	Share patient-level folds. Start with the paired cohort.
Late-fusion method	Owner TBD	Assign an owner. Prespecify a simple fusion baseline.
Export metadata	All model owners	Fix class mapping, attribution scales and score metadata.
Demo and acceptance	Paulo + Junhao Reviewer TBD	Choose demo mode. Verify A30 output and review cases.

H&E and extra mIF batches also need a scope decision before inclusion.

Completion checkpoints

Checkpoint	Evidence of completion
Contract agreement	The team records the endpoint, IDs, splits and model exports.
A30 smoke experiment	Paulo generates local Nemotron output from the synthetic cases.
Real prediction handoff	Model owners supply held-out clinical, mIF and fused scores.
Review and comparison	The team inspects paired metrics and representative narratives.
Demo integration	Junhao displays each patient’s scores, evidence and review status.

Ready now: mock experiment, HPC runner and integration interfaces. Pending: real GPU and model handoffs.